

Virtual Resection Planning using Bezier Surface Interactions in Collaborative VR Environments

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ABSTRACT

The use of virtual reality (VR) has been proposed for collaborative planning or training scenarios to aid surgeons in preparing surgical interventions. In this paper, we explore the usage of cubic Bezier surfaces and their respective interactions in collaborative VR environments to define atypical resection surfaces for planning abdominal organ tumor resections. We conducted a pilot study ($n=10$) to evaluate the usability of the Bezier surface interaction compared to a free deformation approach using the example of liver surgery planning. Moreover, we showed the prototype to an expert and conducted an interview to collect clinical feedback. The results show potential benefits of both interaction techniques that could be essential for virtual resection planning and training. The expert highlighted that combining both techniques could speed up the workflow and provide various options for planning.

Index Terms: 1.3.7 [COMPUTER GRAPHICS]: Three-Dimensional Graphics and Realism—Virtual reality;

1 INTRODUCTION

Surgical resection is a common treatment against tumor-related cancer [1]. However, it is a challenging intervention due to the complex internal structures, such as vascular trees and biliary ducts, and significant anatomical variations between patients [1]. Thus, it requires extensive preoperative planning for each patient. In liver surgery, surgeons define virtual resections according to the patient's imaging data. They take into account accessibility, feasibility, safety distance to all detected cancerous tissues and supply areas of the major vessel structures, and the resulting remaining functional volume of the liver [2].

Preoperative planning is based on a stack of medical volume data representing image slices of the organ acquired from magnetic resonance imaging (MRI) or computerized tomography (CT). While 2D images of CT and MRI data provide anatomical details, 3D visualizations offer a better overview for large-scale features and general configuration [3]. For all desktop-based solutions, depth perception and 3D interactions are challenging [4]. Several studies show that immersive VR could provide a better environment for planning [5–7].

Collaborative VR allows multiple users to join in the same shared virtual environment. It has been used for training communication and collaboration among interprofessional teams [8,9]. Surgeons can exchange information, cooperate, and share decision-making in VR. Furthermore, the use of collaborative VR in the context of training medical students allows them to safely perform virtual resections, improve their skills respectively, and discuss the results with the training supervisor directly in the virtual environment. Results

of previous work show potential benefits of collaborative VR for medical training and planning [8, 10].

In this work, we explore the use of a different interaction approach with a Bezier surface for abdominal organ resection planning. We integrated this technique into, and enhanced it with, the collaborative VR environment. We conducted a pilot study ($n=10$) to assess the usability of the technique in comparison with a deformable cutting plane (DCP) approach [10] using the example of liver surgery planning. Moreover, an interview was carried out with a domain expert to collect clinical feedback and provide insights regarding the advantages and disadvantages of both interaction techniques.

2 RELATED WORK

3D visualization of the organ offers various options to define the risk structures and to perform a virtual resection. A simple solution for resection planning is the drawing of lines in 2D views of the slices, which are then transformed into a 3D resection surface based on a shape-based interpolation view [11]. However, it is time-consuming to draw in many slices, even with the reduction to a subset based on interpolation. Mostly desktop-based solutions were developed, e.g., deformable cutting plane, spline-based resection, and reconstructions from lines drawn in 2D to interactively define the resection surface [11–13]. The improved shape and depth perception of virtual reality (VR) supports the understanding of internal structures and enhanced planning and training [14, 15].

Palomar et al. [13] make use of Bezier surfaces to define a custom resection surface. Another interaction technique for resection planning is performed by erasing parts of a 3D visualization model [16]. The eraser could have various geometrical shapes like wedges, clip planes, cylinders, and spheres. However, this approach was considered too difficult to perform with the required precision. An alternative technique was proposed using of a deformable cutting surface, which can be deformed by molding it with a certain strength in an interaction point and a cosine-based surrounding area [12].

Hattab et al. [17] investigated the utility of VR for spatial understanding in surgery planning. Boedecker et al. [9] evaluated a VR system for preoperative liver surgery planning. The results show that using VR was perceived as easier compared to desktop-based surgical planning systems. With advances in 3D printing, even physical visualizations have been proposed to further support the preoperative planning [18]. The results of a preclinical study comparing 3D PDFs, 3D printed models, and VR from Huettl et al. [18] reveal that the 3D printed models and VR models enable a better and slightly faster anatomical orientation compared to the visualization on 3D PDFs.

In recent years, collaborative VR has been an essential topic for medical training and planning. Chheang et al. [10, 19] presented VR environments that offer simultaneous collaboration and interactions ranging from preoperative planning to operating room training. Fischer et al. [20] presented a VR system for volumetric medical image visualization that supports collaborative interactions. Their results suggest that collaborative VR represents a unique team discussion and training environment.

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Figure 1: System overview for virtual resection planning: (a) users join the VR environment and explore the patient cases, and (b) the resection surface with Bezier surface interaction. The blue spheres mark the interactive control points. The yellow lines on the liver models surface were used to define the initial position. The risk map with the distance to the tumor is visualized on the Bezier surface.

3 SYSTEM IMPLEMENTATION

We used the *Unity* game engine for system development and *Photon networking* to provide load-balancing service for shared network sessions and data synchronization. HTC Vive Pro and its components were used for VR setup. In the following sections, we explain the prototype's features with a combination of the interaction technique of the Bezier surface (see Fig. 1).

3.1 Bezier Surface

A Bezier surface describes a parametric surface via a set of control points, which can be considered as points of attraction that pull the surface towards them. The advantage of the Bezier surface is that major deformations of wide areas can be achieved with a few control points rather than deforming manually. In addition, the mathematical model of the Bezier surface guarantees a level of smoothness depending on the chosen degree of the polynomial. Similarly to Palomar et al. [13], we chose a Bezier surface of 3rd degree in both directions with 16 control points. While a higher degree with more control points would offer more precision, it would slow the performance, as more interactions on control points would be needed in general.

3.2 Bezier Surface Interaction

The major challenge with the Bezier surface approach is designing the interaction such that it is feasible for the user. It also requires the necessary tools to modify it swiftly. The controllers are represented with virtual hands in VR. The general interactions were implemented to interact with the UI, grab objects, and work with pointers.

Interaction with the Bezier surface is selecting and deselecting one or more control points and moving them. The position of a control point is visualized with a small blue sphere. If currently selected, the sphere is colored yellow. The first method to select control points is to interact with them with a pointer. To support this, they are highlighted if aimed at, and the hitbox for the pointer is larger than the actual sphere to make aiming easier but avoid too much visual occlusion.

A control point is selected if it collides with the index finger of the virtual hand. A two seconds period, in which this interaction cannot be repeated, is added to avoid accidentally deselecting it again at the very next moment. For this interaction, the hitbox of the index finger is slightly enlarged to reduce the precision requirements when trying to select multiple points with one single waving motion of the hand.

The last option of selection is the UI (see Fig. 1b). The 16 control points may be selected via buttons. Moreover, control points can be grouped. The currently selected control points are emphasized by coloring their respective buttons. All can be either selected or deselected at once. Single rows and columns of the grid and the four inner control points can be selected as groups since they are often moved together. These selectors behave as toggles to allow quick switching between complementary sets of points. Furthermore, a single selection mode can be activated.

Once the user has made the selection, all currently selected control points can be moved by holding the trigger button on the controller. For as long as it is held down, the control points will copy the relative movement of the virtual hand, and the Bezier resection surface will deform accordingly in real time. Notice that for this movement, it is not necessary to have the virtual hand close to the control points or the liver model to avoid visual occluding with the hand and accidentally selecting control points.

4 EVALUATION

To assess the usefulness of the interaction technique, we showed our prototype to a domain expert and collected clinical feedback. Moreover, a pilot study was conducted and compared to the DCP technique. The DCP technique was implemented based on [10].

Expert's Feedback To acquire valid positions of the resection surfaces that should act as targets for the study participants, we asked a general and visceral surgical resident to use our prototype and perform the virtual resection on four patient cases. The session was conducted remotely in the collaborative VR environment. The expert joined in VR from the Johannes Gutenberg University of Mainz, Germany while the investigator joined from the Otto-von-Guericke University of Magdeburg, Germany. The expert tested both interaction techniques, i.e., Bezier surface interaction and deformable cutting plane [10]. The final configurations of these resection surfaces were saved. After that, we conducted a semi-structured interview to collect clinical feedback regarding the prototype's usefulness, advantages, and disadvantages.

Pilot Study For the pilot user study, we recruited ten participants, primarily from our department. Most of them had a background in computer science and medical engineering. Two of them were female, and three were left-handed. All participants had previous VR experience: seven classified themselves as experienced users, and three reported having only used it a few times before. A minority of the three were said to be familiar with the concept of Bezier curves or surfaces.

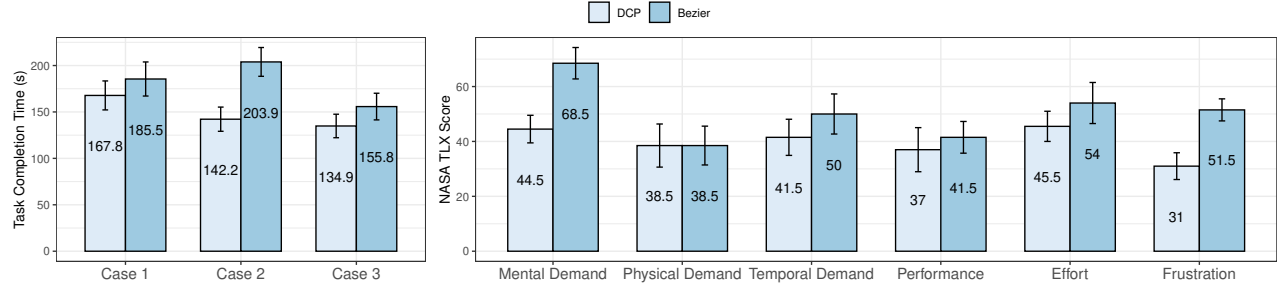


Figure 2: Results of the pilot study: (left) task completion time and (right) NASA TLX questionnaire results.

The study procedure was divided into several steps. First, the participants were asked to answer the demographic questionnaire, and techniques and study procedures were explained. The participant was then asked to explore the prototype and perform the virtual resection with both interaction techniques on the model designated for training. Four patient cases resulting from the expert's planning were used for this pilot study, one was used in training, and the other three were used for tasks, i.e., case 1, case 2, and case 3. The participants were connected together with the investigator in the VR environment for the training session. However, we asked the participants to perform the virtual resection planning without assistance in the actual session. During the training, all necessary controls were explained and demonstrated by a researcher who joined the same VR session. Task completion time was measured for the actual tasks, from the moment the interactive surface is spawned until the pressing of the *Finish*-button. After completion of the tasks, the participants were asked to answer the *NASA Task Load Index* (TLX) questionnaire [21] for both interaction techniques. Lastly, the participant was asked for additional remarks and feedback.

5 RESULTS

In the following, we describe the results of the pilot study, including participant performance and feedback.

5.1 User Performance

The average completion time (s) for all tasks were ($M = 148.30$, $SD = 44.56$) for the DCP and ($M = 181.73$, $SD = 53.26$) for the Bezier technique. Descriptive results of user performance are shown in Table 1 (see also Fig. 2). We further analyzed the data with the pairwise t test. There was a significant difference ($t = -3.58$, $df = 9$, $p < 0.006$) of task completion time on Case 2. The results show that the task completion time for DCP was less than the Bezier technique. However, there were slight differences for other tasks. According to the feedback, three participants reported that they struggled with either the concept or the practical application of the Bezier surface to deform the resection surface. They needed more guidance during the training and took more time.

Table 1: Descriptive results of task completion time (s).

Technique	Case 1	Case 2	Case 3
DCP	167.8 (49.33)	142.2 (41.22)	134.9 (40.03)
Bezier	185.5 (58.02)	203.9 (49.05)	155.8 (45.38)

All entities are in the format: mean value (standard deviation).

5.2 Questionnaire Results

The overall unweighted NASA-TLX rating (Raw TLX) was ($M = 39.67$, $SD = 20.10$) for the DCP and ($M = 50.67$, $SD = 21.54$) for the Bezier surface interaction. The questionnaire contains six categories:

mental demand, physical demand, temporal demand, performance, effort, and frustration (see Fig. 2). The results of pairwise t test show significant differences on mental demand ($t = -4.20$, $df = 9$, $p < 0.002$) and frustration ($t = -3.99$, $df = 9$, $p < 0.003$). The DCP was relatively easy to use for those who are not familiar with Bezier curves. However, the physical demand was relatively similar between both techniques.

5.3 Participant Feedback

The expert and participant feedback was collected, including suggestions on possible changes and features that could enhance and further improve the prototype.

DCP Interaction Technique For the DCP approach, the feedback was mainly on the orientation and direction of the deformation from the controller rather than from the resection surface. Other feedback discussed visualizing the risk map on which the deformation is applied. All users showed similar strategies: all of them used/employed similar changing deformation directions, most adjusted the rate of deformation (force) to a higher value, and some adjusted the radius to the area they were currently working on, while the rest instead worked with a small radius but strong force. A few remarks were made concerning the controls of the deformation, mostly requesting that the deformation direction be oriented from the controller instead of from the target.

Bezier Surface Interaction There were plenty of suggestions for the Bezier approach, starting with activating the UI from a longer distance or binding more UI functions to shortcuts with controller buttons. For the accidental selection, disabling the selection via collision during the movement of points or sound feedback on selection was suggested.

For deformation based on the Bezier technique, many different interaction strategies were observed. The most common strategies involved selecting the outer or inner control points via the group selection options, arranging them, and then continuing in a single selection mode. The direct selection of single points via UI was also frequently used, but mainly for deselecting points. The participants placed the outer control points directly on the intersection between the shadow target surface and the liver model, forming a rather small surface and extending it into the model with the inner control points. Aside from these preferences, some participants strongly favored one of the selection approaches, e.g., virtual hand collision. One of the often-observable hindrances was the accidental selection of control points via collision with the virtual hand, which happened to some degree to all participants; it took a lot of time to correct the selection, or even the accidentally-made deformations, when the wrong selection was not noticed soon enough. This often happened during the movement of control points due to the tendency of most participants to keep their virtual hands close to the control point, although this was unnecessary.

Another issue was the arrangement of the points whose positions had been swapped in their position, leading to convolutions or other unexpected surface behavior. Once that happened, the users had to spend a long time solving it, mostly by comparing via the synchronized highlighting of points on the model and UI to the position the point in question originally belonged. Most participants who faced this problem in the early tasks noticed it and actively tried to avoid it later. Participants also suggested that displaying numbers next to the points or showing small indicators of their original positions on the surface could provide a better understanding of their arrangement.

A shrinking feature that moves the selected control points radially towards or from their shared center of mass was often requested as it would be helpful for the strategies developed by the participants. One suggestion suitable for both techniques was to define the initial position of the deformable plane differently by simply freely translating, rotating, and scaling it instead of drawing the circle on the liver model. Moreover, the freeform navigation requires a significant amount of learning for the users [22]. Therefore, a feature with properties to improve the learning experience of freeform navigation should be investigated in VR.

6 CONCLUSION

We have presented a new approach for defining atypical resection surfaces for planning virtual resections, making use of cubic Bezier surfaces in a collaborative VR environment. The results of the pilot study show insights for both interaction techniques. However, the number of participants was relatively small, and the aim of the formative study was to test and collect feedback for further improvement since it was an early stage of development.

Many participants reported needing more training for the Bezier surface interaction. Also, the Bezier method is complicated at the beginning, as confirmed by the NASA TLX questionnaire ratings. However, participants stated that familiarity with the Bezier technique could lead to better results and faster deformation. The DCP, in its simplicity, makes it easy to learn but also leaves little opportunity for improvement or adding additional features. For the Bezier technique, however, there is plenty of room for improvement.

The results show small differences. The Bezier technique has potential to quickly deform the resection surface, while the DCP technique could provide a detailed deformation, especially to avoid affected vascular structures. The expert and participants confirmed that combining both interaction techniques could be an option for resection planning. Moreover, participant feedback for both techniques indicates potential to improve the workflow. Future work aims to conduct an extensive clinical evaluation, supplement with additional cases, and transfer to other surgical disciplines, e.g. kidney and pancreatic surgery.

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